Start Assignment

- Due Monday by 11:59pm
- Points 15
- Submitting a text entry box or a file upload
- Attempts 0
- Allowed Attempts 2
- Available until Mar 17 at 11:59pm

Module 8: Research Project

Programming Language Research Project

During this week's Canvas Discussion, we learned about various programmers and coders. This week we begin our semester research project about programming languages used/invented by those very programmers and coders we studied. Each student will **select a different high-level programming language** to research, write a formal research paper, and present your research findings to the entire class. The following is a list of the top 10 programming languages according to **Northeastern University's Align program (https://graduate.northeastern.edu/knowledge-hub/most-popular-programming-languages/)**. **Other programming languages (https://en.wikipedia.org/wiki/List_of_programming_languages)** are available upon prior request and approval by *~Dr. H*.

Northeastern U.'s Top 10 Programming Languages

1. Structured Language Query (SQL) - **Raymond**
2. Python
3. Java
4. JavaScript - **Caitlin**
5. C# (ISO/IEC 23270) - **Hung**
6. C++ (ISO/IEC 14882)
7. R - **Raissa**
8. C (ISO/IEC 9899) - **Jesus**
9. Go - **Kush**
10. Perl - **Nick**

Below is another list Programming Languages for your consideration

1. Ada
2. Assembly - **Sean**
3. COBOL (ISO/IEC 1989) - **Mercy**
4. Fortran (ISO/IEC 1539)
5. IBM RPG
6. Lisp (ISO/IEC 13816) - **Jordyn**
7. PHP
8. Ruby
9. Rust
10. SmallTalk
11. Swift (Apple programming language) - **Hassan**
12. TypeScript - **Daria**
13. B Language - **Issam**
14. Julia language - **Walker**
15. There are hundreds of others! Please send Dr. H. an email requesting permission if you are interested in a programming language not listed.

Your Semester Research Project

Your research project will be written and submitted for grading in four (4) phases. General requirements include the following.

- Write a **five (5) page** research paper and no more or less that discusses your approved programming language. Use a 12-point font and double spacing. Your research paper needs to contain a combination of **ten (10) trade journal and peer-reviewed, scholarly references,** properly cited and formatted according version 7 of the APA-style guidelines. Always remember to reference AI writing assistance, if used. Also keep in mind the famous quote by Blaise Pascal, the French philosopher, inventor, and mathematician. In his Lettres Provinciales, **Blaise Pascal** (https://iep.utm.edu/pascal-b/) famously wrote: "***I would have written a shorter letter, but I did not have the time***."
- Include a separate cover page with the title, your name, the date, the name of the course, and your Professor's name
- Include a 100-word abstract with keywords.
- Include the 10 requirements listed below in the Phase I deliverable. If you have space left in your five page, you are welcome to include anything else of interest about your programming language.
- Include a reference list on a separate page at the very end of your research paper with at least a combination of ten (10) peer-reviewed and trade journal references.

- Properly format your research paper according to version 7 of the [APA style guidelines, (https://guides.rider.edu/citations/apa)](https://guides.rider.edu/citations/apa) including in-text citations and the reference list. Also, if used, reference any AI writing assistance also according to APA-style guidelines.
- Correct spelling and grammar are important, and count toward your overall grade for your research project.

Phased Deliverables:

1. **Phase I:** Be prepared to make a preliminary presentation (5-minutes) about your research during class on **Wed, March 18, 2026**. Your presentation should include:
   (1) the name of the programming language you selected;
   (2) the inventor(s) of the language;
   (3) the year it was invented;
   (4) is it considered a high-level language and/or a general purpose language or something else identified;
   (5) is it a compiled or interpreted language or both; (6) what are the built-in security features;
   (7) main purpose/use of the language;
   (8) is there an AI coding assistant, such as GitHub Copilot, Gemini Code Assist, Claude, etc., that you would recommend for use with your programming language;
   (9) is the programming language backed by one of the ISO standards or an other standard; and
   (10) the reason for your choice/what interested you about the language.

   Also include at least 4 or 5 references, which can be a combination of trade journals and scholarly articles. By **11:59 pm ET on Mon, March 16, 2026**, please upload your **presentation slide deck through this assignment link**.

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2. **Phase II:** Submit and present your 100-word abstract with keywords of your research paper during class on **Wed, March 25, 2026 by 11:5 PM ET**. There will be a **separate assignment link in Canvas** to submit your abstract with keywords.

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3. **Phase III:** Submit your completed research paper through Canvas by **Mon, March 30, 2026 by 11:59 pm ET.** There will be a **separate assignment link in Canvas** to submit your research paper.

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4. **Phase IV:** Present the key findings of your research paper during class on **Wed, April 1, 2026**. You will have 5 minutes to present and ~2 minutes for follow up discussion. There will be a s**eparate assignment link in Canvas** to submit your presentation slides.

Remember assignments may be submitted only one day late and will incur a 10% late penalty.

Please let me know if you have any questions at anytime - **e.hawthorne@northeastern.edu (mailto:e.hawthorne@northeastern.edu)**
